

Homework 8

Due: Jun. 15th

Turn in your homework in class

Rules

- Please work on your own. Discussion is permissible, but extremely similar submissions will be judged as plagiarism.
- Please show all intermediate steps: a correct solution without an explanation will get zero credit.
- Please submit on time. No late submission will be accepted.
- Please prepare your submission in English only. No Chinese submission will be accepted.
- Please retain 3 decimal places if rounding is required (unless otherwise specified).

1

[12 points]

Find the Laplace transform of each of the following functions.

(a)

$$f(t) = 20e^{-500(t-10)}u(t-10)$$

(b)

$$\begin{aligned} f(t) = & 5t[u(t) - u(t-2)] \\ & - (5t-20)[u(t-2) - u(t-6)] \\ & + (5t-40)[u(t-6) - u(t-8)] \end{aligned}$$

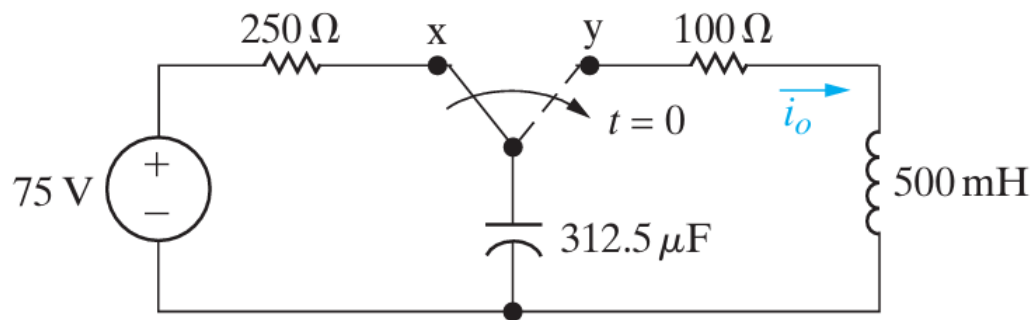
(c) Find the inverse Laplace transform:

$$F(s) = \frac{s^3 + 2s^2 + s + 1}{(s+1)(s+9)(s^2 + 6s + 13)}$$

2

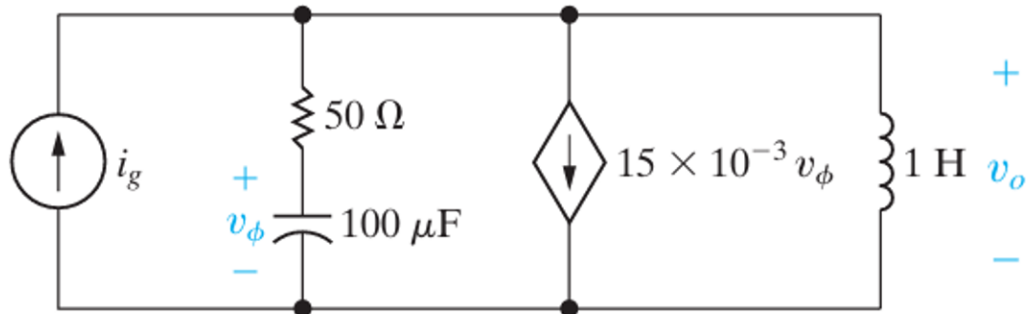
[15 points] The switch in the circuit shown in figure has been in position x for a long time. At $t = 0$, the switch moves instantaneously to position y.

- (a) Construct an s -domain circuit for $t > 0$.
- (b) Find $I_o(s)$, the current in the Laplace domain.
- (c) Find $i_o(t)$, the current in the time domain.



3

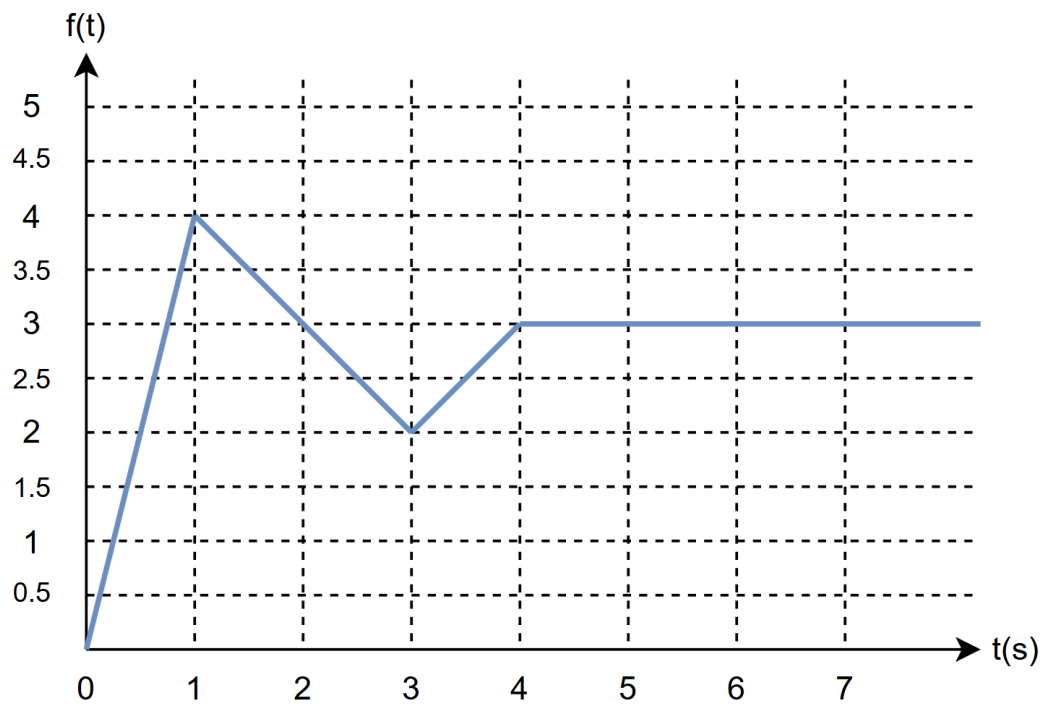
[10 points] Find v_o in the circuit shown in figure if $i_g = 20u(t)$ mA. There is no energy stored in the circuit at $t = 0$.



4

[15 points] The function $f(t)$ is shown below. $f(t) = 0$ for $t \leq 0s$ and $f(t) = 3$ for $t \geq 4s$.

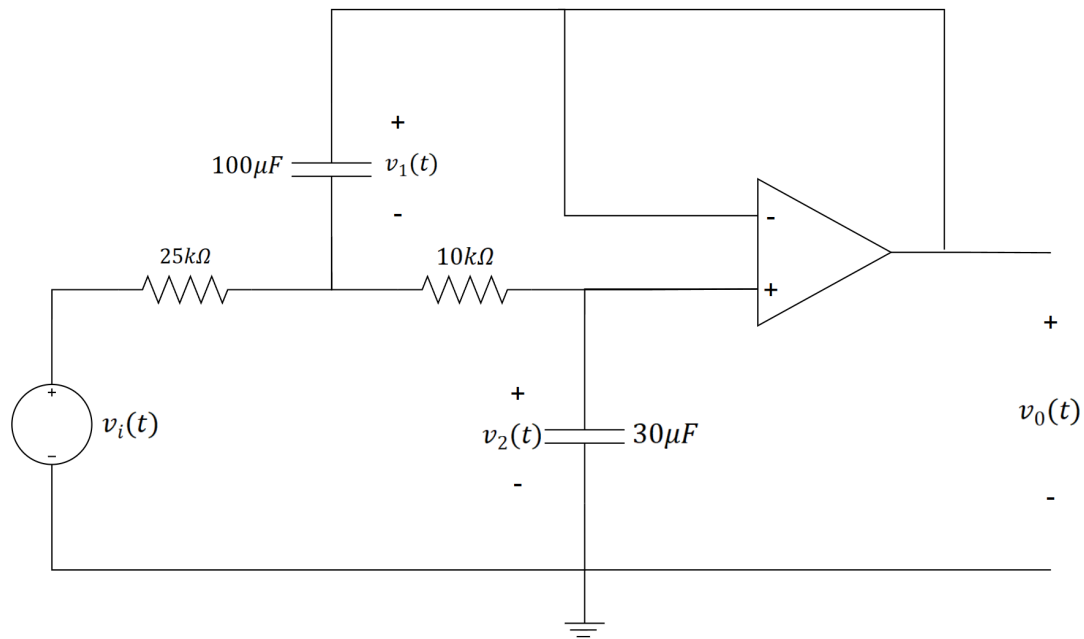
- (a) Calculate the Laplace transform of $f(t)$.
- (b) Verify the initial value theorem of $f(t)$.
- (c) Verify the final value theorem of $f(t)$.



5

[16 points] The circuit is shown below. The ideal operator amplifier is working at its linear region and $v_1(0^-) = 0V, v_2(0^-) = 0V$. Find:

- the transfer function $H(s) = \frac{V_o(s)}{V_i(s)}$.
- the impulse response $h(t) = L^{-1}\{H(s)\}$.
- the response $v_o(t)$ when $v_i(t) = e^{-2t}u(t)V$.
- the response $v_o(t)$ when $v_i(t) = 8\cos(2t)V$.

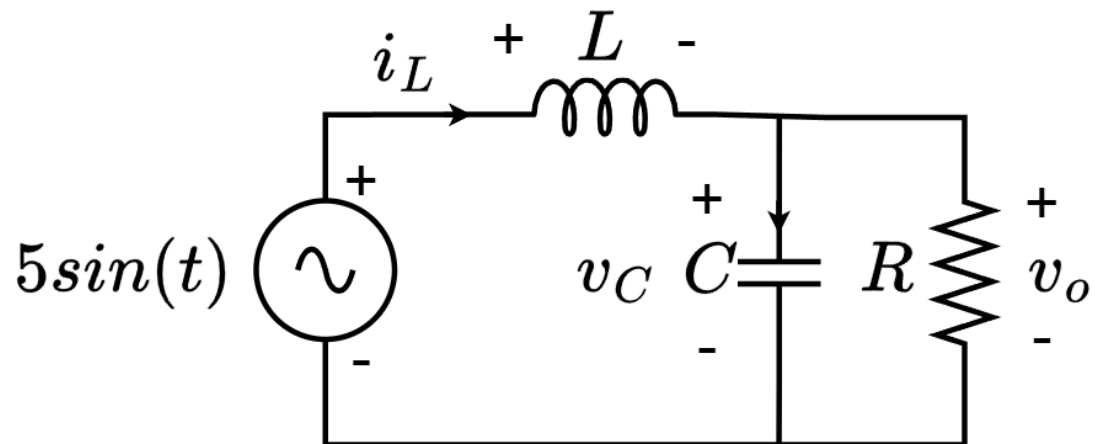


6

[16 points] Known that $L = 0.5 \text{ H}$, $C = 1 \text{ F}$, $R = 0.5 \Omega$, $v_C(0^-) = 1 \text{ V}$, $i_L(0^-) = 1 \text{ A}$.

(a). Use time domain method to determine $v_o(t)$ for $t > 0 \text{ s}$.

(b). Use Laplace transform to determine $v_o(t)$ for $t > 0 \text{ s}$.



7

[16 points] Background: MOSFET (metal oxide semiconductor field effect transistor) is an important device in modern integrated circuits and computer architectures. A digital integrated circuit may contain millions of MOSFETs, but the area is only on the millimeter order. In this question, you need to solve a problem of a circuit with only one MOS.

Here is a common source stage amplifier. There are several capacitances between gate, source and drain, which are the three terminals of a MOS. Although we do not need to consider them in DC analysis, they are crucial in AC analysis. Now we need to consider these capacitances in the Laplace domain.

Parameters: R_S means the resistance of the voltage source, R_D means the resistance of drain of the MOS, C_{GS} means the capacitance between gate and source, C_{GD} means the capacitance between gate and drain, C_{DB} means the capacitance between drain and body, and g_m is called transconductance whose unit is $S = \Omega^{-1}$. All these parameters are already known.

Based on this information, derive the transfer function. The equivalent circuit is shown below.

$$H(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{(C_{GD}s - g_m)R_D}{R_S R_D \xi s^2 + [R_S(1 + g_m R_D)C_{GD} + R_S C_{GS} + R_D(C_{GD} + C_{DB})]s + 1}$$

$$\xi = C_{GS}C_{GD} + C_{GS}C_{DB} + C_{GD}C_{DB}$$

